Palatial Pools

Technical Disclosure Statement

Palatial Pool Services Pty Ltd
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1. Purpose

This Technical Disclosure Statement outlines the technology, tools, and methodologies employed by Palatial Pool Services in the delivery of its digital services, diagnostic tools, and customer-facing platforms. This document is provided for transparency and to inform customers and stakeholders about how our systems operate.

2. Platform Overview

Palatial Pools operates a comprehensive digital platform that includes a suite of interconnected services designed to enhance the pool care experience.

Our primary interface is the Website and Service Portal at `palatialpools.com.au`. This web-based platform provides information about our pool services, hosts online tools, and facilitates customer interaction. Integrated into this portal is the Pool Photo Diagnostics system, an advanced image analysis tool that assesses pool conditions

from photographs. This system is custom-trained on thousands of real-world diagnostic situations sourced from hundreds of cities globally.

For accurate service estimation, we utilize a Satellite Pool Volume Calculator. This mapping tool leverages satellite imagery, computer vision, and geometric calculations to automatically detect and estimate pool volume. Service delivery is managed through Palatial Link, our customer management and scheduling system. Finally, we employ Dustin, a privately trained diagnostic assistant that provides real-time guidance and product recommendations, developed entirely in-house.

3. Diagnostic Intelligence Architecture

3.1 Training Foundation

The Palatial Pools diagnostic engine represents a significant investment in applied artificial intelligence. It has been custom-trained on an extensive corpus of real-world diagnostic situations, providing a breadth of exposure that exceeds the capacity of any individual technician.

Category	Coverage Details
Water Conditions	Algae blooms, chemical imbalances, turbidity, discolouration
Surface Issues	Staining, etching, calcium deposits, tile damage
Equipment States	Filter conditions, pump issues, chlorinator faults
Environmental Factors	Seasonal variations, UV exposure effects, debris patterns
Geographic Diversity	Tropical, temperate, arid, and coastal pool environments

3.2 Infrastructure: L40 Server-Class Computing

To ensure real-time diagnostic inference, Palatial Pools operates dedicated high-performance computing infrastructure hosted in Sydney, Australia. This L40 Server-Class Platform is purpose-built for our specific workloads.

The system is powered by four NVIDIA A100 80GB Tensor Core GPUs, providing a total of 320 GB of GPU memory. These are interconnected via an NVLink 600 GB/s interface to ensure rapid data transfer. The compute layer is supported by dual AMD EPYC 7763 processors, offering 128 cores, alongside 2 TB of DDR4 ECC system memory and a redundant NVMe storage array.

3.3 Inference Pipeline

The diagnostic process follows a multi-stage inference pipeline optimized for both accuracy and speed.

1. **Image Pre-Processing**: Uploaded images undergo automatic exposure and white balance correction, resolution normalization, noise reduction, and metadata extraction.
2. **Multi-Head Attention Analysis**: The system performs parallel analysis across multiple dimensions, including water color spectrum analysis, edge detection for surface conditions, and pattern recognition for biological growth.
3. **Diagnostic Output Generation**: The final stage produces a confidence-weighted condition assessment, a severity classification (Immediate, Monitor, or Normal), actionable recommendations, and product matching from verified databases.

> **Performance Specifications:** Our inference pipeline achieves a latency of less than 3 seconds per image and can handle over 200 simultaneous analyses, backed by a 99.9% uptime SLA.

4. Satellite Pool Volume Calculator

4.1 Automatic Pool Detection

The Volume Calculator incorporates computer vision to automatically detect pool boundaries from satellite imagery.

The process begins with address geocoding, converting the user's input into precise coordinates. High-resolution satellite imagery is then retrieved. This image is processed through our inference pipeline to identify pool boundaries, which are subsequently converted to geographic coordinates. A fluorescent polygon overlay is drawn on the interactive map, and the surface area is calculated with a confidence score.

4.2 Calculation Method

Volume is calculated using the standard formula: `Volume (litres) = Surface Area (m²) × Average Depth (m) × 1000`. The surface area is computed using the Spherical Law of Cosines via the Google Maps Geometry library, accounting for the Earth's curvature at the measurement scale.

While the auto-detection provides a percentage confidence score, users can manually refine detected boundaries. Results are estimates with a typical accuracy range of ±10 to 15%.

5. Technology Stack

5.1 Frontend and Backend Technologies

The customer-facing website is built using modern web technologies, including React, HTML5, and CSS3, ensuring a responsive design optimized for all devices.

The backend infrastructure relies on an Nginx reverse proxy with SSL/TLS encryption. The application layer utilizes Python (Flask) and Node.js microservices. Structured

data storage for customer records and service history is maintained securely, and a CDN optimizes the delivery of static assets.

5.2 Third-Party Integrations

We integrate with select third-party services to provide comprehensive functionality. Google Maps is utilized for satellite imagery, geocoding, and area calculation, which involves sharing address input and map interactions. Payment processors handle service payment collection securely (PCI-compliant), and email services manage customer communications.

6. Data Privacy and Security

6.1 Data Collection and Security Measures

Palatial Pools collects address information for the Volume Calculator, pool photographs for diagnostics, contact information via forms, and anonymized usage analytics.

Security is paramount. All data transmission uses TLS 1.2+ encryption. API keys and credentials are stored server-side only. We do not share customer data with third parties for marketing purposes. Uploaded images are processed in real-time and are not permanently stored. Our server infrastructure includes robust firewall protection and regular security updates.

6.2 Compliance

Palatial Pools operates in accordance with the Australian Privacy Act 1988 and the Australian Privacy Principles (APPs). Customers may request access to, correction of, or deletion of their personal information at any time.

7. Service Delivery Technology

Service scheduling is managed through the Palatial Link platform, which coordinates technician availability, service history, equipment inventory, and customer communication. Professional water testing during service visits utilizes calibrated equipment, with digital records maintained for trend analysis and proactive maintenance.

8. Intellectual Property and Disclaimers

All custom software, designs, branding, and content on the Palatial Pools platform are the intellectual property of Palatial Pool Services Pty Ltd unless otherwise attributed.

Please note that our digital tools provide estimates and indicative results only. They do not replace professional pool inspection and water testing. Palatial Pools is not liable for decisions made solely based on digital tool outputs. Service recommendations and chemical dosage calculations should always be confirmed by a qualified pool technician.

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